

# Mega7 Simulation Analysis Report

## 0. Executive summary

Mega7 is the **deepest balance sweep to date** and the first to use the new pooled `ratings_combined.csv` artefact (every battle from every stage and weather in one flat table). It spans **nine team sizes (2v2–10v10)** × **six weathers**, 360 pieces (120 bases × 3 levels), with a combined median of **237 matches/piece** — ~20× the per-cell depth of mega6.

Five headline findings:

1. **Faction balance is sound.** Champion vs enemy win rate sits at 0.507 / 0.491 pooled and stays within  $\pm 0.02$  at every team size. Every pathology below is *\*within\** the roster, not a champion/enemy asymmetry.
2. **The mage deficit persists and is a small-format problem.** Mage is the weakest role pooled (win rate 0.458 vs non-mage 0.515; gap +0.057). The within-tier deficit is worst at 2v2 (0.079), shrinks through the mid sizes, and *\*inverts\** at 7v7–8v8 (mages slightly ahead). Same shape as mega6, marginally worse at 2v2.
3. **Weather favor is modest and working correctly — ~+0.01, not the +0.18 a first pass suggested.** An earlier read of this dataset reported a dominant +0.18 own-vs-counter swing; that was a **metric artefact** (the `counter_weather_wr` column zero-fills weathers a piece never played — 37% of rows, including every **clear**-affinity piece — and averaging those zeros craters the counter mean). Recomputed from raw per-weather win rates, the own-weather edge is **+0.010 own-vs-counter** and **+0.008 own-vs-clear-baseline**, positive for *\*every\** affinity (no snow/thunder inversion — that too was thin-slice noise). The favor system is applied and behaves as designed; the metric has been fixed (§6).
4. **T10 hybrid boss kits under-deliver vs their power budget — again.** Hybrid has the *\*highest\** raw win rate (0.559) but the *\*most negative\** pooled `wr_delta` (−0.030): Aurion (clear T10 L3, −0.245), Aerion (thunder T10 L3, −0.165), Storm Tyrant (−0.089). They win because they are high-power, but less than their power says they should. This recurs from mega6.
5. **Level premium is steep but roughly on-budget.** L3 pooled win rate 0.593 vs L1 0.424; L3 `wr_delta` −0.021 (slightly over-delivers). By design, not flagged.

One real watch-item: **timeouts climb to 44% at 10v10**, compressing role signal at the largest sizes.

**On how odd the `wr_deltas` actually are (§8, expanded this revision):** the pooled-residual distribution is near-normal and only mildly heavy-tailed — most pieces are within  $\pm 0.10$  of budget, and the named outliers are real ~2 tail events within their tier (`wd_z` column), not a fat-tailed mess. Crucially, the residuals are *\*not\** context-volatile: subtracting the binomial-noise floor leaves **only 8 of 360 pieces with any across-context spread beyond sampling noise**, and those excesses are tiny ( $\leq 0.056$ ) and thin-sample. So the actionable signal is the **absolute** outliers (§8.2), not inconsistency. Boxplots (by role + tier), histogram/Q-Q, and a context-volatility ranking are in §8.

## 1. Dataset and method

| Field                | Value                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Directory            | results/mega/mega7/                                                                                                                   |
| Run date             | 2026-06-03                                                                                                                            |
| Engine state         | current working tree: poison percentage-decay (V.25 decay_fraction=0.2), barrier system, Glade Heron rework, weather favor $\pm 0.30$ |
| Stages               | 2v2, 3v3, 4v4, 5v5, 6v6, 7v7, 8v8, 9v9, 10v10 (no 1v1)                                                                                |
| Per-stage rated rows | 19,200                                                                                                                                |
| Combined pieces      | 360 (120 bases $\times$ 3 levels)                                                                                                     |
| Weathers             | clear, cloudy, mist, rain, snow, thunder                                                                                              |
| max-ticks            | 12,000 (engine default — sudden death engaged)                                                                                        |
| Baseline             | mega6 (cache_mega6.rds; raw dir deleted)                                                                                              |

**The combined file is the canonical aggregate.** Per-(stage,weather) ratings still exist for slicing, but per-cell samples are thin (median 3–6 matches). `ratings_combined.csv` pools every battle — pair-game weighted — so its per-piece win rates and cross-weather metrics are the reliable read (median 237 matches). All §7–§8 outlier work uses it.

**wr\_delta interpretation (carries from prior reports).** `expected_wr` is the deterministic power-threshold model: 1.0 if your team's  $\Sigma P$  exceeds the opponent's, 0.0 if less, 0.5 if equal, averaged over the actual opponent field. `wr_delta = win_rate - expected_wr`. Near 0 = on-budget; negative = kit under-delivers vs power; positive = over-delivers.

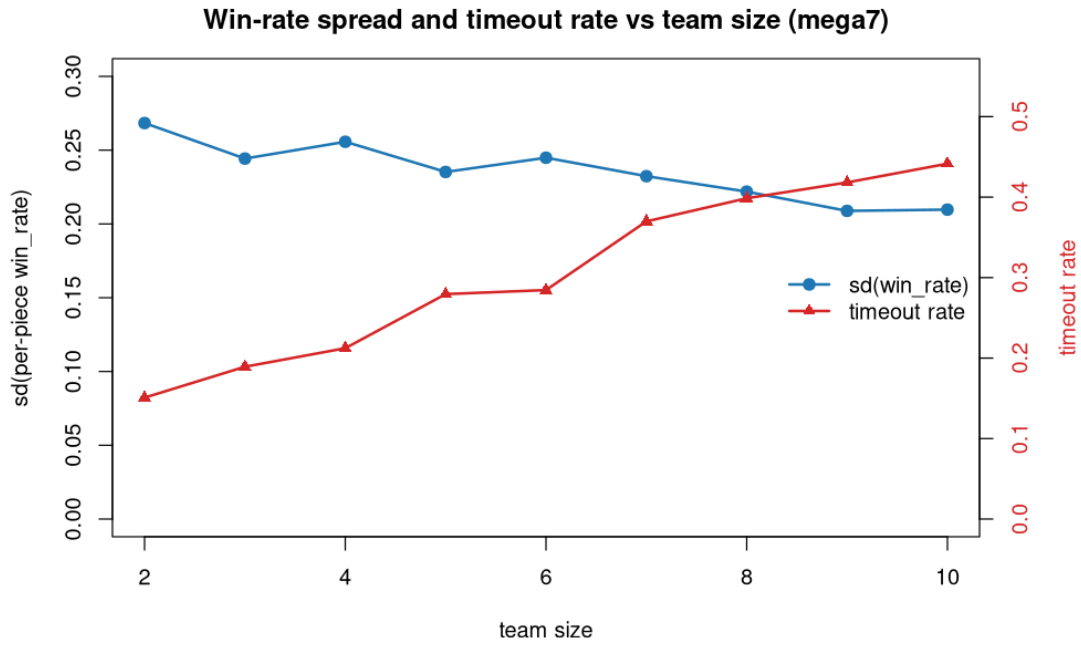
**Provenance caveat.** Mega7 was generated from the current uncommitted working tree. Kit-level reads for pieces touched this session (Glade Heron, poison users, Hierarchy) are indicative; a clean post-commit re-run is advisable before acting on those specific rows.

## 2. Aggregate balance

| stage | champion | enemy | sd(win_rate) | timeout |
|-------|----------|-------|--------------|---------|
| 2v2   | 0.500    | 0.505 | 0.268        | 0.151   |
| 3v3   | 0.505    | 0.492 | 0.244        | 0.189   |
| 4v4   | 0.501    | 0.491 | 0.256        | 0.212   |
| 5v5   | 0.489    | 0.493 | 0.235        | 0.279   |
| 6v6   | 0.502    | 0.503 | 0.245        | 0.285   |
| 7v7   | 0.508    | 0.475 | 0.232        | 0.370   |
| 8v8   | 0.519    | 0.486 | 0.222        | 0.399   |
| 9v9   | 0.513    | 0.489 | 0.209        | 0.418   |
| 10v10 | 0.504    | 0.485 | 0.210        | 0.442   |

Champion/enemy parity holds across all formats. Win-rate spread declines only modestly with team size (0.27  $\rightarrow$  0.21), much less than mega6's collapse (0.23  $\rightarrow$  0.09). The cause is **per-stage sampling depth**, not a real effect: mega7's per-(piece, stage) samples are thin (median 3–6 matches — the `--total-battles` budget spread across nine sizes  $\times$  six weathers), so each per-piece win rate carries large binomial noise that inflates sd at every size. mega6 reached 50–222 matches/piece at its top sizes, so its sd genuinely collapsed. Read mega7's \*pooled\* (combined) win rates for per-piece signal; treat per-stage sd as noise-dominated.

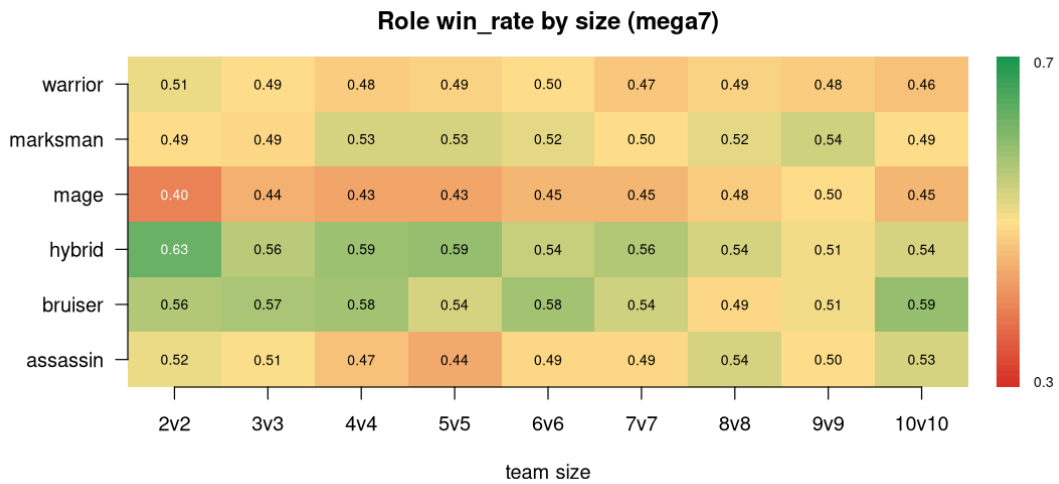
**Timeouts are a rising concern:** 15% at 2v2  $\rightarrow$  44% at 10v10. Every timeout is a draw, pulling win rates toward 0.50 and compressing differences at the top sizes.



win\{}\_rate spread and timeout vs team size

### 3. Role balance

#### 3.1 Win rate by team size



Role win\{}\_rate by size (heatmap)

Pooled (combined) role win rates:

| role     | win_rate |
|----------|----------|
| mage     | 0.458    |
| warrior  | 0.486    |
| assassin | 0.496    |
| marksman | 0.515    |
| bruiser  | 0.546    |
| hybrid   | 0.559    |

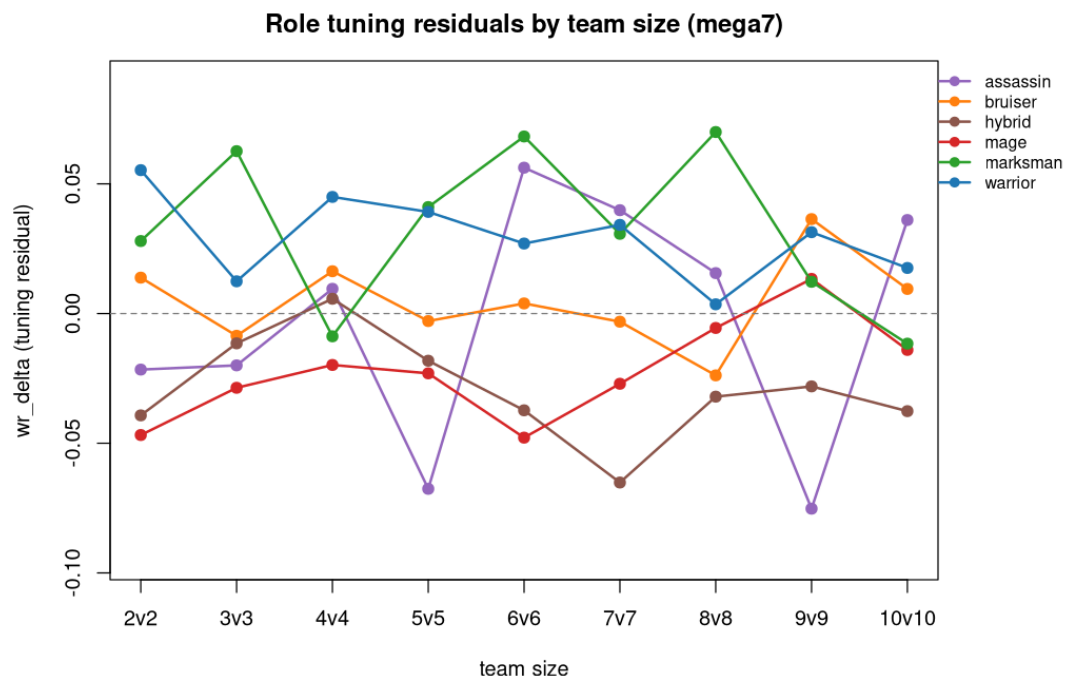
Mage is the floor at every size except 9v9. Hybrid and bruiser are the ceiling — the same two outliers flagged in mega6.

### 3.2 Tuning residuals (wr\_delta)

Pooled (combined) role `wr_delta`:

| role     | wr_delta |
|----------|----------|
| hybrid   | −0.030   |
| mage     | −0.018   |
| assassin | −0.001   |
| bruiser  | +0.008   |
| warrior  | +0.024   |
| marksman | +0.029   |

The story splits from raw win rate. **\*\*Hybrid is the most \*under-budget\* role despite the highest raw win rate — its members are high-power T10 bosses that win on stats but under-deliver on kit.** Marksman and warrior over-deliver\*\* (+0.025–0.029): they punch above their power budget. Mage is under on both raw and budget.

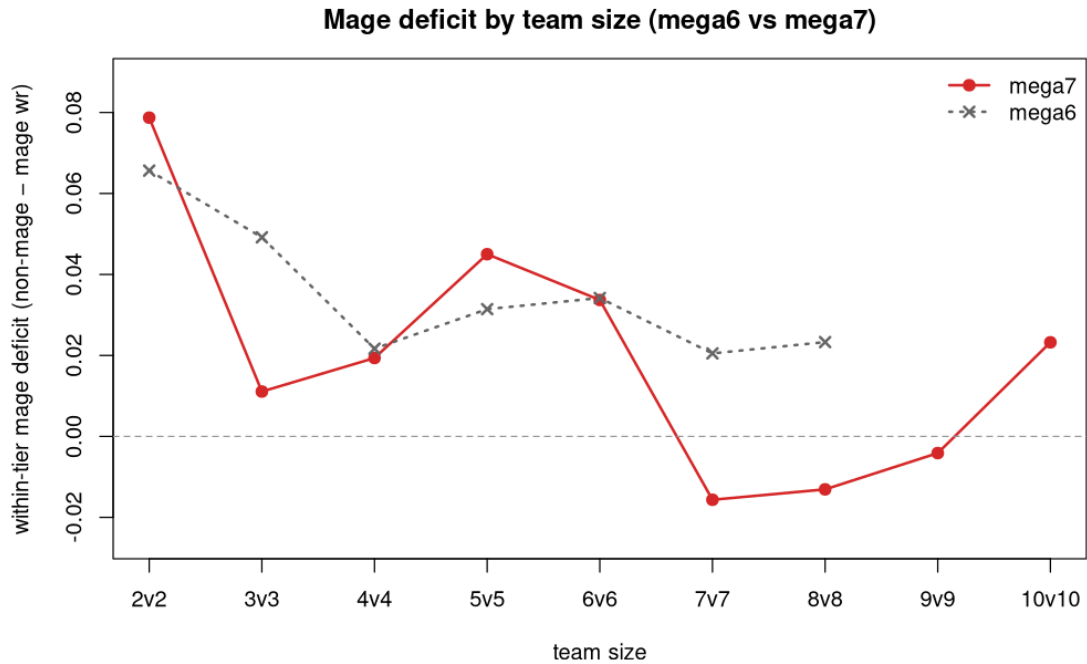


Role tuning residuals by team size

#### 4. Mage deficit

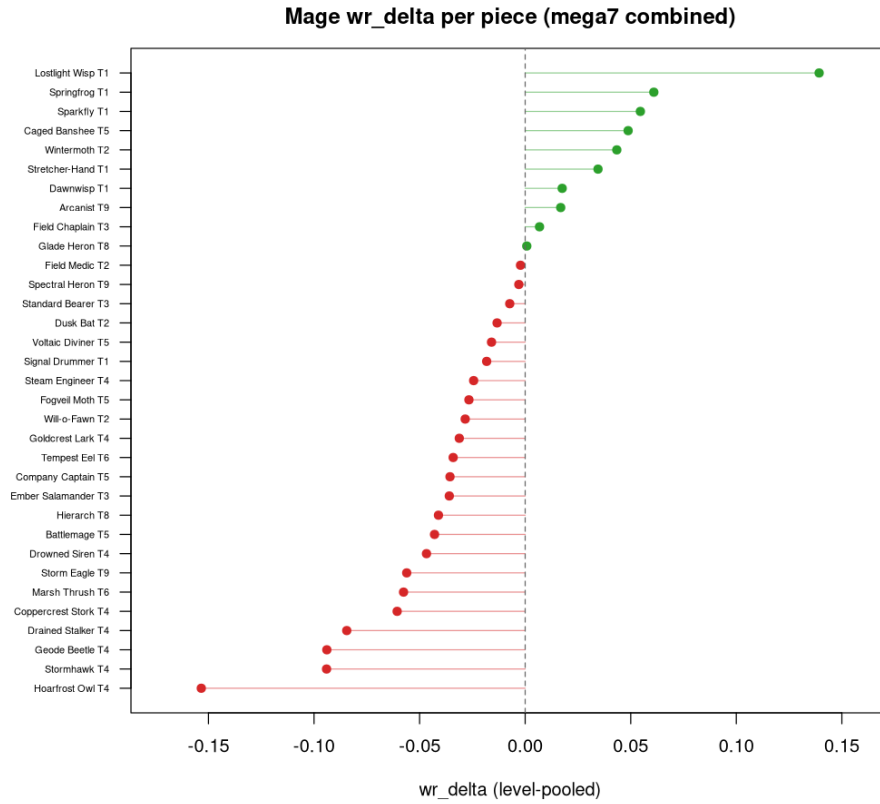
| stage | mage  | nonmage | within_tier_def | cor(tier,wr) |
|-------|-------|---------|-----------------|--------------|
| 2v2   | 0.398 | 0.542   | 0.079           | 0.383        |
| 3v3   | 0.444 | 0.519   | 0.011           | 0.290        |
| 4v4   | 0.434 | 0.519   | 0.019           | 0.322        |
| 5v5   | 0.432 | 0.514   | 0.045           | 0.255        |
| 6v6   | 0.454 | 0.521   | 0.034           | 0.154        |
| 7v7   | 0.453 | 0.506   | −0.016          | 0.265        |
| 8v8   | 0.480 | 0.511   | −0.013          | 0.204        |
| 9v9   | 0.499 | 0.502   | −0.004          | 0.223        |
| 10v10 | 0.452 | 0.510   | 0.023           | 0.223        |

The within-tier deficit is sharply concentrated at **2v2 (0.079)** and is gone or inverted by 7v7–8v8. Mages scale \*into\* relevance with team size — they need allies to survive long enough to ramp. Versus mega6 at matched sizes the picture is mixed: 2v2 slightly worse (+0.013), but 3v3/7v7/8v8 notably better (−0.036 to −0.038).



Mage deficit by team size: mega6 vs mega7

The worst individual mages (pooled `wr_delta`) are **Hierarch** (clear T8 L3, −0.221), **Hoarfrost Owl** (snow T4 L3, −0.184), **Standard Bearer** (clear T3 L2, −0.179), **Fogveil Moth** (mist T5 L3, −0.170), **Drowned Siren** (rain T4 L1, −0.165). Note Hierarch — it received the on-death barrier this session, which does nothing for its own survival; it remains the single most under-tuned mage. **Glade Heron L3 is on-budget (+0.022)** despite a 0.903 raw win rate (T8 L3 is \*expected\* to win that much) — the rework landed in the right place.

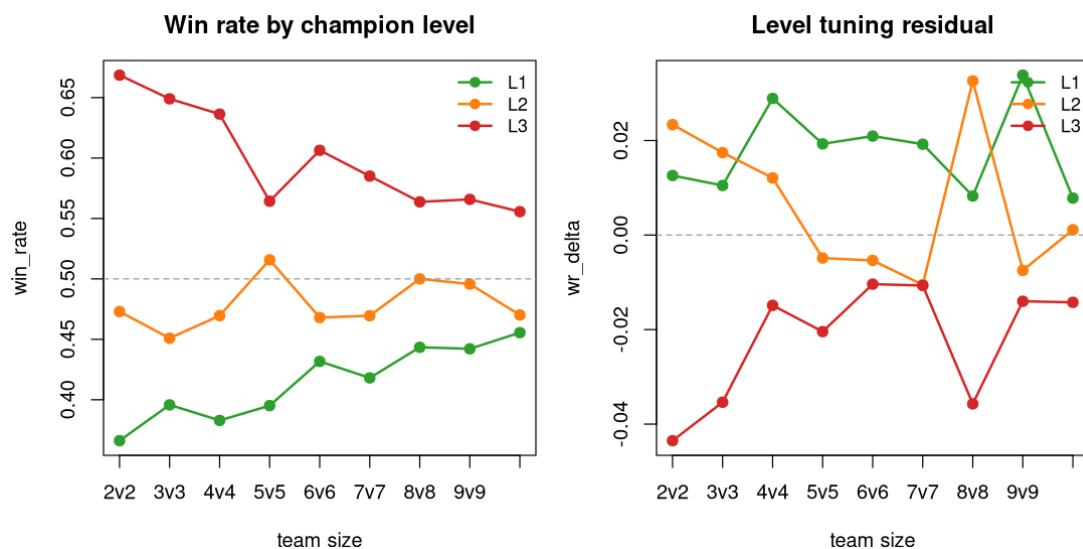


Mage wr\{\}\_delta per piece (combined)

## 5. Level effects

| level | win_rate | wr_delta | timeout |
|-------|----------|----------|---------|
| 1     | 0.424    | +0.019   | 0.279   |
| 2     | 0.480    | +0.004   | 0.277   |
| 3     | 0.593    | -0.021   | 0.304   |

The level premium is steep (L1→L3 = +0.17 raw) but **wr\_delta** stays small: L1 slightly under-delivers, L3 slightly over. Mixed-level teams will be L3-dominated, which is by design. No action needed.



Win rate and tuning residual by champion level

## 6. Weather effects

**Headline correction.** A first pass on this dataset reported a dominant **+0.18** own-vs-counter weather advantage. That number was wrong — a metric artefact, not a balance fact. This section uses the corrected computation (raw per-weather win rates); the favor effect is real but **modest** ( $\sim +0.01$ ).

**The bug.** The `counter_weather_wr` column defaults to **0.0** when a piece has no games in its counter weather. That conflates "no data" with "0% win rate". It fires constantly: a piece's counter weather is one specific weather of six, and every `clear`-affinity piece has *\*no\** counter weather at all (NEUTRAL on the ring). Result — `counter_weather_wr == 0.0` in **37.1%** of rows. Averaging those zeros in collapses the mean counter and manufactures a fake gap:

| computation                               | own          | counter      | own — counter |
|-------------------------------------------|--------------|--------------|---------------|
| buggy column-average (zeros in)           | 0.505        | 0.332        | +0.173        |
| <b>correct (raw per-weather win rate)</b> | <b>0.509</b> | <b>0.499</b> | <b>+0.010</b> |

Fixed in code: `weather_metrics` now emits `NaN` (not 0.0) for a weather the piece never played, and the CSV writer leaves the cell empty so aggregators skip it. This report recomputes from raw per-weather win rates and is unaffected.

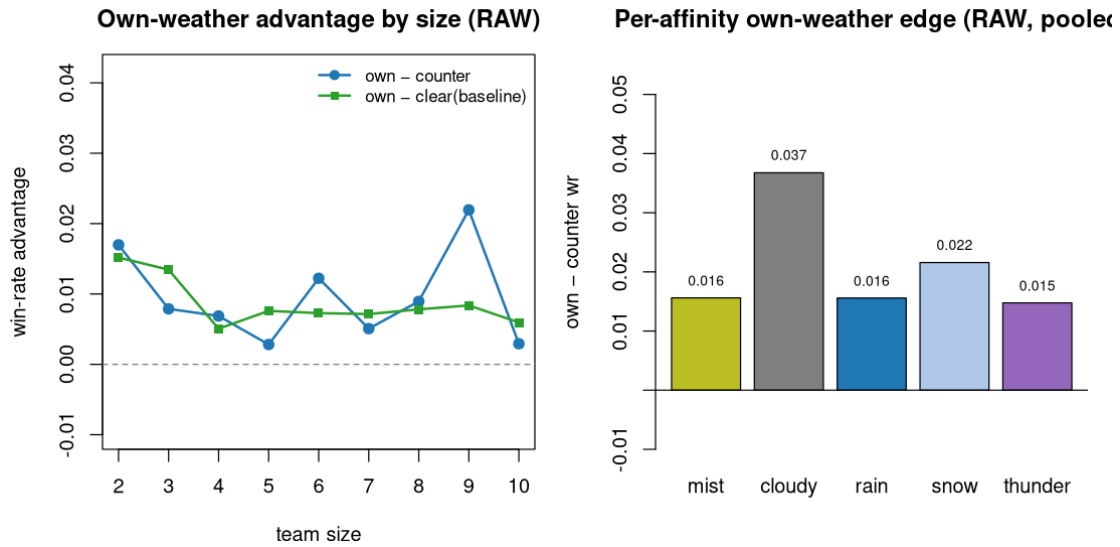
**Corrected own-weather advantage by size** (own = win rate in own affinity weather; clear = no-favor baseline; counter = weather that preys on you):

| stage | clear (base) | own   | counter | own — counter | own — clear |
|-------|--------------|-------|---------|---------------|-------------|
| 2v2   | 0.500        | 0.515 | 0.498   | +0.017        | +0.015      |
| 4v4   | 0.500        | 0.505 | 0.498   | +0.007        | +0.005      |
| 6v6   | 0.500        | 0.507 | 0.495   | +0.012        | +0.007      |
| 8v8   | 0.500        | 0.508 | 0.499   | +0.009        | +0.008      |
| 10v10 | 0.500        | 0.506 | 0.503   | +0.003        | +0.006      |

**Corrected per-affinity** (raw, pooled all stages):

| affinity | clear | own   | counter | own — clear | own — counter |
|----------|-------|-------|---------|-------------|---------------|
| mist     | 0.512 | 0.513 | 0.498   | +0.002      | +0.016        |
| cloudy   | 0.506 | 0.530 | 0.494   | +0.025      | +0.037        |
| rain     | 0.515 | 0.530 | 0.515   | +0.016      | +0.016        |
| snow     | 0.509 | 0.516 | 0.495   | +0.007      | +0.022        |
| thunder  | 0.491 | 0.511 | 0.496   | +0.020      | +0.015        |

Every affinity gains in its own weather and loses (or is flat) in its counter weather — **the favor system works as designed, and the earlier snow/thunder “inversion” was thin-slice noise**, not a real effect. The magnitude is small (single-digit win-rate points), so weather favor is a flavour-and-edge system, \*not\* a dominant axis — it does not overshadow kit or role. Whether  $\pm 0.30$  favor magnitude should produce a \*larger\* effect than this is a separate design question, but it is not overpowered.



Weather: corrected own-advantage by size and per-affinity edge

## 7. Timeouts

| role     | 2v2   | 5v5   | 8v8   | 10v10 |
|----------|-------|-------|-------|-------|
| assassin | 0.131 | 0.312 | 0.352 | 0.483 |
| bruiser  | 0.320 | 0.378 | 0.508 | 0.398 |
| hybrid   | 0.120 | 0.289 | 0.350 | 0.405 |
| mage     | 0.160 | 0.248 | 0.406 | 0.448 |
| marksman | 0.078 | 0.257 | 0.379 | 0.404 |
| warrior  | 0.145 | 0.271 | 0.412 | 0.469 |

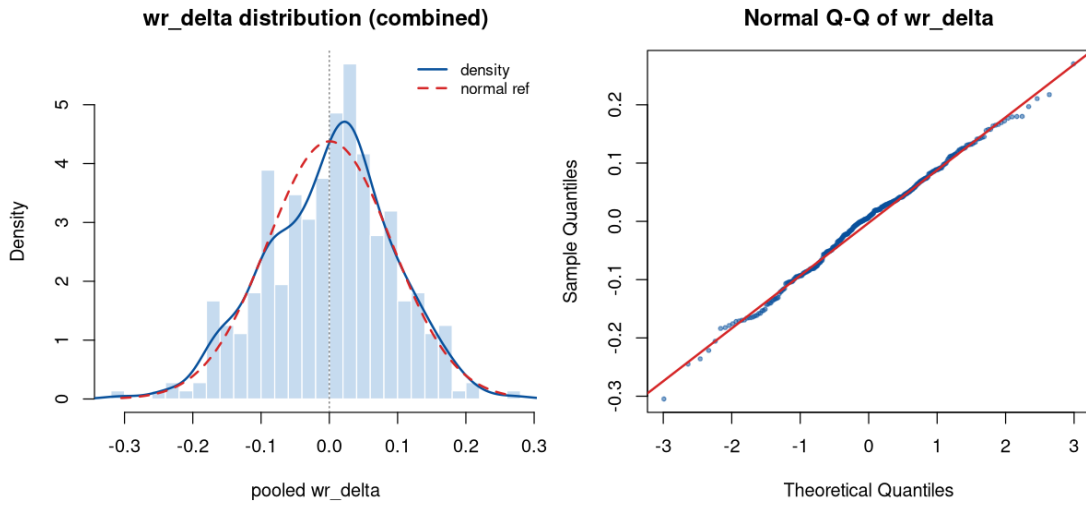
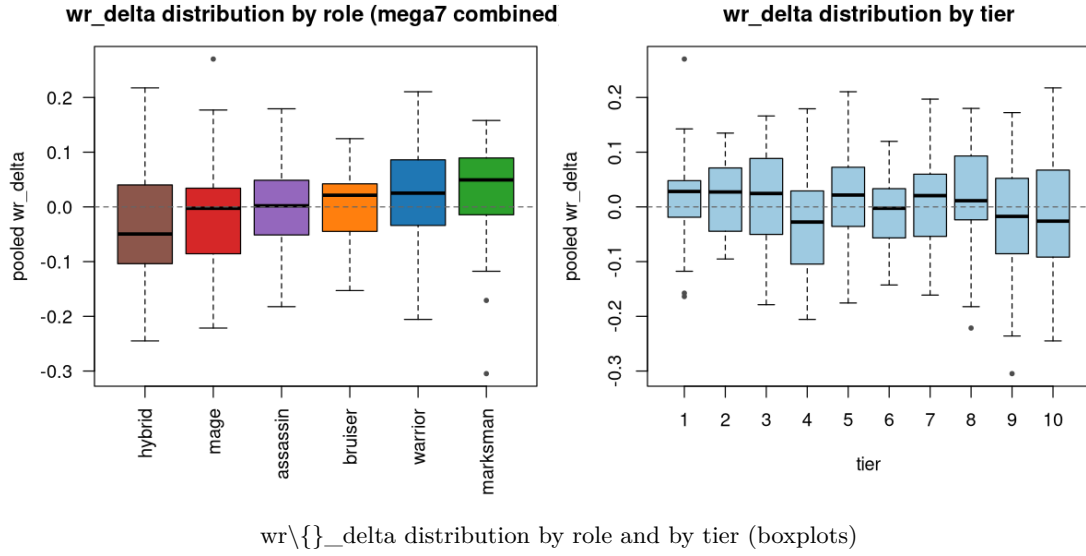
Bruisers stall hardest at small/mid sizes (high survivability, low burst), consistent with mega6. At the largest sizes nearly half of all battles time out — the 12,000-tick cap is too tight for 10v10 to resolve organically. Large-format role reads should be treated cautiously because of this.

## 8. Outliers and tuning priorities

### 8.1 How odd are the residuals? Distribution shape

Before naming individual pieces, the question “how odd are these `wr_deltas`” is best answered by the \*shape\* of the pooled-residual distribution. The boxplots below break `wr_delta` out by role and by tier; the histogram/Q-Q

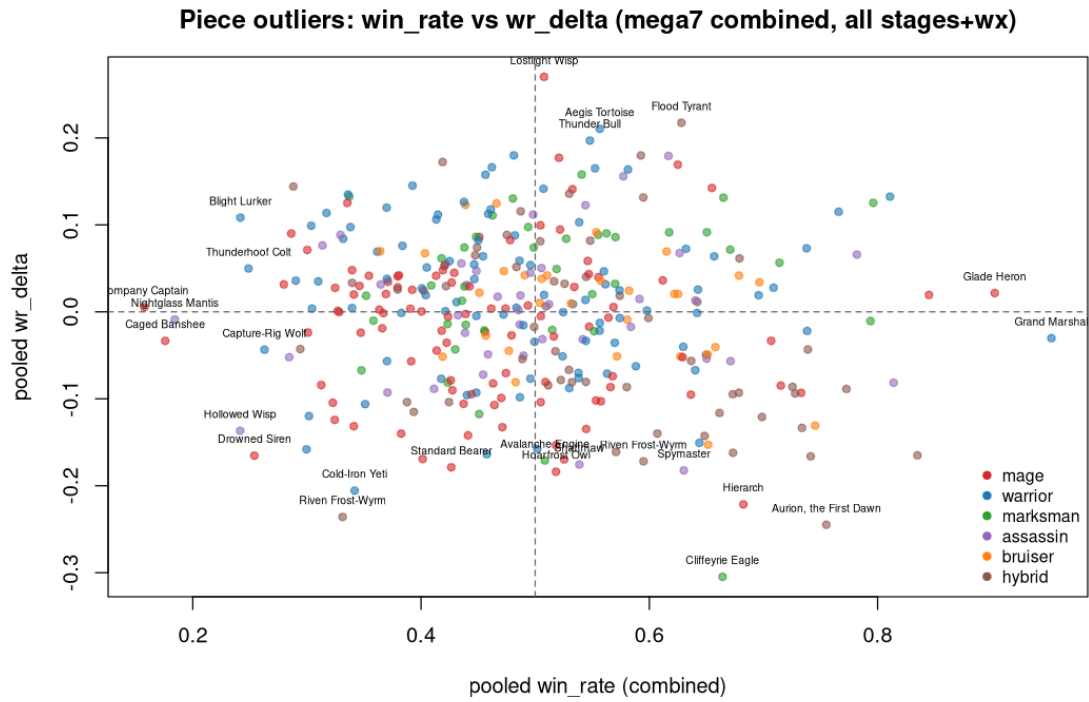
pair shows the overall distribution against a normal reference.



The pooled `wr_delta` distribution is **near-normal in the body and only mildly heavy-tailed** (Q-Q is straight from roughly  $-2$  to  $+2$ , with a handful of points pulling away at each end). Practically: the residuals are *\*not\** pathologically wild — most pieces sit within  $\pm 0.10$  of budget, and the named outliers below are genuine tail events ( $|z| \geq 2$  within their tier), not an artefact of a fat-tailed mess. By role, **hybrid and mage carry the most negative-skewed boxes** (medians below 0, long lower whiskers); **warrior and marksman skew positive**. By tier the spread is fairly uniform — high tiers are *\*not\** systematically noisier once pooled — which is what justifies the within-tier z-score below.

## 8.2 Absolute outliers (largest $|\text{wr\_delta}|$ )

`wd_z` = within-tier z-score of `wr_delta` (how many SD from the piece's *\*tier cohort\** mean).  $|z| \geq 2 \approx$  a genuine per-tier outlier, not tier-position noise.



Most under-tuned (pooled wr\_delta):

| name                          | affinity | role     | tier | level | win_rate | wr_delta | wd_z  |
|-------------------------------|----------|----------|------|-------|----------|----------|-------|
| Cliffeyrie Eagle              | cloudy   | marksman | 9    | 3     | 0.664    | -0.305   | -2.76 |
| Aurion, the First Dawn        | clear    | hybrid   | 10   | 3     | 0.755    | -0.245   | -2.21 |
| Riven Frost-Wyrm              | snow     | hybrid   | 9    | 1     | 0.331    | -0.236   | -2.10 |
| Hierarch                      | clear    | mage     | 8    | 3     | 0.682    | -0.221   | -2.35 |
| Cold-Iron Yeti                | snow     | warrior  | 4    | 2     | 0.342    | -0.206   | -1.87 |
| Hoarfrost Owl                 | snow     | mage     | 4    | 3     | 0.518    | -0.184   | -1.64 |
| Szymaster                     | clear    | assassin | 8    | 3     | 0.630    | -0.182   | -1.96 |
| Standard Bearer               | clear    | mage     | 3    | 2     | 0.426    | -0.179   | -2.19 |
| Shaftmaw                      | cloudy   | assassin | 5    | 2     | 0.539    | -0.176   | -1.96 |
| Avalanche Engine              | snow     | marksman | 5    | 2     | 0.508    | -0.171   | -1.91 |
| Fogveil Moth                  | mist     | mage     | 5    | 3     | 0.525    | -0.170   | -1.90 |
| Umbral, the Mountain's Shadow | cloudy   | hybrid   | 10   | 3     | 0.741    | -0.166   | -1.45 |
| Drowned Siren                 | rain     | mage     | 4    | 1     | 0.254    | -0.165   | -1.44 |

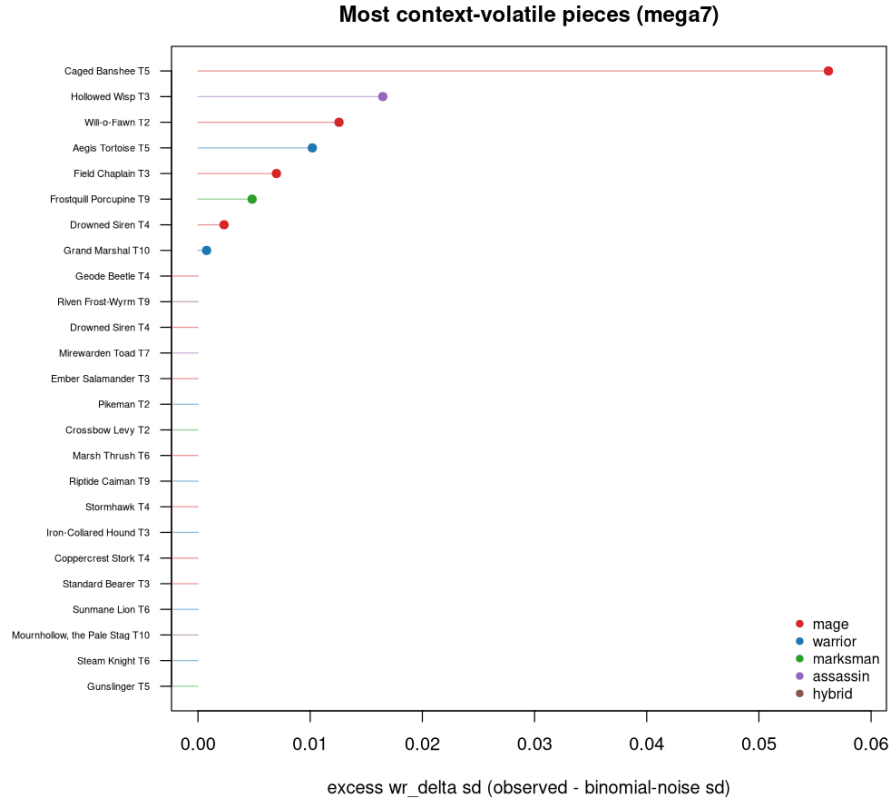
Most over-tuned:

| name                   | affinity | role     | tier | level | win_rate | wr_delta | wd_z  |
|------------------------|----------|----------|------|-------|----------|----------|-------|
| Lostlight Wisp         | mist     | mage     | 1    | 1     | 0.508    | +0.270   | +2.89 |
| Flood Tyrant           | rain     | hybrid   | 10   | 2     | 0.628    | +0.217   | +2.25 |
| Aegis Tortoise         | clear    | warrior  | 5    | 3     | 0.557    | +0.210   | +2.05 |
| Thunder Bull           | thunder  | warrior  | 7    | 2     | 0.548    | +0.197   | +2.20 |
| Storm Tyrant           | thunder  | hybrid   | 10   | 2     | 0.593    | +0.180   | +1.88 |
| Frostfang<br>Wolverine | snow     | warrior  | 8    | 1     | 0.481    | +0.180   | +1.67 |
| Hollow Elk             | mist     | assassin | 4    | 3     | 0.617    | +0.179   | +2.23 |
| Caged Banshee          | mist     | mage     | 5    | 3     | 0.521    | +0.177   | +1.70 |
| Sundered Lord          | mist     | hybrid   | 9    | 1     | 0.419    | +0.172   | +1.84 |
| Voltaic Diviner        | thunder  | mage     | 5    | 2     | 0.625    | +0.169   | +1.62 |
| Heavy Knight           | clear    | warrior  | 4    | 2     | 0.581    | +0.164   | +2.07 |
| Iron-Collared<br>Hound | snow     | warrior  | 3    | 3     | 0.462    | +0.166   | +1.70 |
| Cannoneer              | clear    | marksman | 8    | 1     | 0.541    | +0.158   | +1.45 |

The under-tuned list is dominated by **high-tier hybrids/mages that win on raw power but miss their budget** (Aurion, Umbra, Hierarch) and **snow pieces** (Riven Frost-Wyrm, Cold-Iron Yeti, Hoarfrost Owl, Avalanche Engine) — reinforcing the snow-affinity anomaly from §6. The over-tuned list is led by a T1 mage (Lostlight Wisp,  $z=+2.9$  — the single oddest piece in the sweep) and the rain/thunder hybrid Tyrants. Full 30-row absolute-outlier table: `tables/m7_abs_outliers.csv`.

### 8.3 Spread outliers — context-volatility of wr\_delta

A piece can also be "odd" by being **inconsistent**: a pooled `wr_delta` near 0 that hides large swings across formats/weathers is as much a balance concern as a flat large residual. Per-(stage,weather) cells are thin here (often 1–2 matches), so raw spread is dominated by binomial noise. We compute each piece's match-weighted `sd(wr_delta)` across its 54 contexts, subtract the **expected binomial-noise sd** (estimated from the piece's pooled win-rate, \*not\* per-cell — a per-cell estimate collapses to 0 on single-match cells and fabricates spread), and rank by the **excess**.



Most context-volatile pieces (excess wr\_delta sd beyond noise)

**Key result: spread is almost entirely sampling noise.** Only **8 of 360** pieces show `excess_sd > 0` — i.e. for ~98% of pieces, the observed across-context `wr_delta` spread is fully explained by thin-sample binomial noise. The genuine context-volatile pieces:

| name                 | affinity | role     | tier | tot_matches | wd_pooled | wd_sd | noise_sd | excess_sd |
|----------------------|----------|----------|------|-------------|-----------|-------|----------|-----------|
| Caged Banshee        | mist     | mage     | 5    | 60          | +0.050    | 0.540 | 0.483    | +0.056    |
| Hollowed Wisp        | mist     | assassin | 3    | 156         | -0.144    | 0.261 | 0.244    | +0.016    |
| Will-o-Fawn          | mist     | mage     | 2    | 330         | -0.038    | 0.225 | 0.212    | +0.013    |
| Aegis Tortoise       | clear    | warrior  | 5    | 84          | +0.143    | 0.401 | 0.391    | +0.010    |
| Field Chaplain       | clear    | mage     | 3    | 390         | -0.055    | 0.197 | 0.190    | +0.007    |
| Frostquill Porcupine | snow     | marksman | 9    | 312         | +0.013    | 0.163 | 0.158    | +0.005    |
| Drowned Siren        | rain     | mage     | 4    | 192         | +0.230    | 0.231 | 0.228    | +0.002    |
| Grand Marshal        | clear    | warrior  | 10   | 198         | -0.015    | 0.117 | 0.116    | +0.001    |

The excesses are **small in absolute terms** ( $\leq 0.056$ ) and the leaders are thin-sample (`tot_matches` 60–390). Three of the eight are **mist** pieces — mist is the affinity whose counter-relationship gets sampled least evenly, so a residual mist-driven swing is plausible but unconfirmed at this depth. **Takeaway:** `wr_delta` is not meaningfully context-volatile in mega7 once noise is removed — the outliers worth acting on are the *absolute* ones in §8.2, not spread. Full table: `tables/m7_spread_outliers.csv`. To confirm any spread finding, re-run with `--total-battles` raised ~5–10× so per-cell depth lifts above the binomial-noise floor.

## 9. Recommendations

- **Mage early-game power (priority).** Target the 2v2 within-tier deficit specifically — it vanishes at scale, so the fix is early survivability/damage, not late-game. Worst offenders: Hierarchy, Hoarfrost Owl, Standard Bearer, Fogveil Moth, Drowned Siren.
- **T10 hybrid boss kits (Aurion, Aerion, Storm Tyrant) under-deliver vs budget** — recurring since mega6. Their kits need a pass; they should top the chart, not merely win on raw power.
- **Hierarchy.** Most under-tuned mage; the new on-death barrier helps allies, not itself. Needs a personal-survivability or damage buff.
- **Weather favor: no action needed.** Corrected analysis (§6) shows it working as designed at a modest  $\sim +0.01$  edge. If a *stronger* competitive weather effect is wanted, the  $\pm 0.30$  magnitude could be revisited — but it is not a balance problem. The metric bug that overstated it has been fixed in `weather_metrics`.
- **Validated this session:** Glade Heron L3 is on-budget ( $+0.022$ ); the poison percentage-decay model produced no runaway poison outliers in team play.

## 10. Reproducibility

```
# 1. Run the sweep (current working tree)
python -m tools.simulation.mega --total-battles <N> --skip 1v1 \
    --workers 16 --out results/mega/mega7

# 2. Analysis + tables + cache
Rscript reviews/mega_sim/11_mega7.R

# 3. Figures
Rscript reviews/mega_sim/12_mega7_plots.R

# 4. This PDF
python3 reviews/mega_sim/build_mega7_pdf.py
```

Tables written to `reviews/mega_sim/tables/m7_*.csv`; cache to `cache_mega7.rds`; figures to `reviews/mega_sim/plots/m7_*.png`. The engine is byte-deterministic — same tree + same flags reproduce the CSVs exactly.